# DDoS Threats in IoT: A Review of Blockchain-Based Defense Strategies


## Abstract

The rapid expansion of the Internet of Things (IoT) has led to substantial progress in smart technologies; however, it has also introduced severe security risks, particularly Distributed Denial of Service (DDoS) attacks. The extensive deployment, inadequate security configurations, and limited processing capabilities of IoT devices render them highly susceptible to botnet-driven intrusions that disrupt critical services and overwhelm networks. Traditional centralized security models often fall short in addressing these threats because of single points of failure and scalability constraints, highlighting the need for decentralized and resilient security frameworks. This review aims to: (1) analyze the evolving DDoS threats in IoT networks and their attack methodologies; (2) evaluate blockchain-based security mechanisms for DDoS mitigation; and (3) identify challenges and propose future research directions for improving blockchain-integrated IoT security solutions. The study categorizes existing defense strategies into decentralized authentication models, blockchain-enabled Intrusion Detection and Prevention Systems (IDS/IPS), and AI-driven anomaly detection techniques. Furthermore, it investigates the role of smart contracts in automation, federated learning for attack identification, and hybrid blockchain architectures to bolster the security of IoT ecosystems. Despite its promise, the integration of blockchain into IoT security encounters various challenges, including scalability concerns, significant computational demands, interoperability issues, and regulatory compliance. This review provides a comprehensive assessment of blockchain-driven defense mechanisms, offering insights into emerging trends, security challenges, and future directions to optimize IoT network resilience against evolving cyber threats.



## Keywords

IoT security, Blockchain, DDoS detection, AI-based security, Decentralized authentication.


## 1. Introduction

The integration of the Internet of Things (IoT) has revolutionized digital ecosystems by enabling seamless communication across devices across several industries, such as healthcare, smart cities, and industrial automation. However, this rapid growth has also brought forth considerable security concerns, particularly in the form of Distributed Denial of Service (DDoS) attacks. DDoS attacks overwhelm IoT networks with malicious traffic, making services inaccessible to legitimate users (Chaudhary & Mishra, 2023; Kadri et al., 2024; Sripriyanka & Mahendran, 2024). Due to the constrained resources of IoT devices and their insufficient security measures, they have emerged as appealing targets for cybercriminals seeking to exploit weaknesses and establish extensive botnets, as shown by the notorious Mirai malware (Almadhor et al., 2024; Arachchige, Murtaza, et al., 2024; Saha et al., 2023). The growing dependence on IoT devices, combined with the complexity of modern cyber threats, necessitates the urgent need for innovative and robust defense strategies to protect IoT ecosystems from the continuously evolving DDoS threat landscape (Patidar & Jain, 2024; Singh & Jain, 2024; Sripriyanka & Mahendran, 2024; Zaidi, 2024).

Blockchain technology has emerged as a viable approach to address security vulnerabilities in the IoT, particularly in relation to DDoS attacks. The decentralized, immutable, and transparent nature of blockchain's ledger system facilitates secure device authentication,

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anomaly detection, and the management of distributed networks, thereby significantly diminishing the risks linked to centralized points of failure (Aguru & Erukala, 2024; Jiang et al., 2024; Katib & Ragab, 2023). The incorporation of machine learning (ML) alongside blockchain has further bolstered security protocols by allowing for the real-time identification and mitigation of unusual traffic patterns. Recent research indicates that blockchain-integrated IDS, when paired with ML algorithms, can effectively filter out malicious traffic and enhance network resilience (Almadhor et al., 2024; Prasuna et al., 2024). These advancements pave the way for a next-generation in security frameworks, wherein blockchain not only thwarts unauthorized access but also adapts dynamically to the changing landscape of cyber threats.

The remainder of the paper is structured as follows: Section 2 discusses the motivation behind this study, Section 3 outlines the methodology used in our literature review, Section 4 provides a theoretical background on IoT security and blockchain technology, Section 5 presents blockchain-based mitigation strategies, Section 6 explores challenges and future research directions.

## 2. Motivation

The expansion of IoT-driven ecosystems has exposed networks to severe cybersecurity threats, particularly DDoS attacks. These attacks exploit IoT devices' limited security capabilities, making them easy targets for botnet recruitment and large-scale disruptions. Recent studies highlight that IoT botnets account for an increasing proportion of DDoS attacks, leading to severe consequences such as service downtime, financial losses, and compromised data integrity (Gelgi et al., 2024; Lekssays et al., 2023; Saha et al., 2023). Traditional centralized security mechanisms struggle to handle modern DDoS attacks due to the distributed nature of IoT networks, making it crucial to explore decentralized solutions such as blockchain-based security frameworks (W. Liu et al., 2024).

Blockchain technology offers unique security features that make it a viable defense against DDoS threats in IoT networks. The immutability, decentralization, and transparency of blockchain eliminate single points of failure and enable trustless, tamper-proof network operations. Several recent studies have demonstrated the integration of blockchain with ML and artificial intelligence (AI) models to enhance DDoS detection and mitigation mechanisms. For example, a blockchain-based operational threat intelligence (OTI-IoT) framework has been proposed to counter multi-vector DDoS attacks using smart contracts and deep learning-powered IDS (Aguru & Erukala, 2024). Such frameworks have shown promising results in real-time attack detection, lower attack response time, and higher attack prevention rates, indicating blockchain's potential as a transformative solution for securing IoT networks against cyber threats.

Despite these advancements, blockchain-based DDoS mitigation in IoT faces challenges, including high computational costs, scalability issues, and integration complexities with legacy IoT infrastructure. However, new lightweight blockchain implementations and hybrid defense models have been developed to address these concerns. The CREDIT detection scheme, for example, utilizes real-time blockchain-based attack tracking combined with deep learning models to mitigate link flooding attacks with greater efficiency than traditional non-blockchain-based methods (Jiang et al., 2024). The growing number of blockchain-integrated security solutions highlights the need for further research to optimize performance, cost-efficiency, and adaptability in real-world IoT environments.

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## 3. Materials and Methodology

### 3.1 Research Approach

This study employs a systematic literature review (SLR) approach to explore blockchain-based defense strategies against DDoS attacks in IoT networks. The review follows Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines to ensure a transparent, reproducible, and structured synthesis of research findings. The primary objectives of this study are:

1. Analyze DDoS threats in IoT networks and their evolving attack methodologies.
2. Evaluate blockchain technology as a defense mechanism against DDoS attacks in IoT.
3. Identify challenges and propose future research directions for blockchain-based DDoS mitigation.

### 3.2 Data Collection Strategy

To ensure high-quality and credible research sources, this study exclusively retrieves research articles from Scopus, the world's largest abstract and citation database of peer-reviewed literature.

#### 3.2.1 Search Strategy

A structured keyword-based search was conducted in Scopus, applying Boolean operators and filters for precision. The search query was designed as follows:

("DDoS mitigation in IoT" OR "Blockchain security IoT" OR "Smart contract DDoS prevention" OR "Decentralized security IoT" OR "Blockchain for DDoS attack mitigation")

AND (LIMIT-TO (PUBYEAR, 2020) OR LIMIT-TO (PUBYEAR, 2025))

AND (LIMIT-TO (LANGUAGE, "English"))

### 3.3 Inclusion and Exclusion Criteria

A rigorous selection process was applied to ensure relevant, high-impact research is included.

#### 3.3.1 Inclusion Criteria

To ensure the highest quality and relevance of the research included in this review, the inclusion criteria focus on selecting Scopus-indexed peer-reviewed journal and conference papers. By restricting the selection to papers published exclusively between 2020 and 2025, this review prioritizes the latest advancements in blockchain-based DDoS mitigation for IoT networks, capturing recent innovations and emerging trends. Additionally, only studies that explicitly propose blockchain-based security solutions for DDoS attack prevention in IoT ecosystems are considered, ensuring that the research remains highly specialized and relevant to the domain of cybersecurity, IoT, and blockchain technology.

#### 3.3.1 Exclusion Criteria

To maintain the credibility and relevance of this review, studies published before 2020 are excluded, unless they are cited for background context to provide historical insights into blockchain-based DDoS mitigation in IoT networks. This exclusion ensures that the analysis remains focused on the latest advancements and emerging technologies rather than outdated or superseded methodologies.

Furthermore, non-peer-reviewed sources, such as white papers, book chapters, and preprints, are excluded to maintain academic integrity and ensure that all included research has

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undergone a formal peer-review process. Studies that do not specifically focus on blockchain-based DDoS mitigation in IoT are also removed to maintain a sharp focus on relevant security solutions. Additionally, duplicate studies or redundant findings from the same authors are excluded to prevent repetitive analyses and biased results, ensuring that the review presents unique, well-rounded, and diverse perspectives on blockchain-enabled cybersecurity for IoT environments.

## 4. Theoretical Foundations

This section presents an overview of the key concepts and technologies related to DDoS attacks in IoT and the role of blockchain-based defense mechanisms. It includes discussions on blockchain technology, smart contracts, DDoS threats in IoT networks, and machine learning-based security models.

### 4.1 DDoS Attacks in IoT

IoT devices, ranging from smart home assistants and cameras to industrial sensors and healthcare monitors, often lack strong security measures. Many of these devices operate with default credentials, weak encryption, and infrequent firmware updates, making them easy targets for cybercriminals. Once compromised, these devices can be turned into a botnet, a network of infected systems remotely controlled by hackers to launch large-scale DDoS attacks. Chaudhary & Mishra (2023) explained that hackers utilize IoT devices known as zombie machines to flood the service with unwanted requests. From the viewpoint of technology suppliers, such as Fortinet (2023) cybercriminals employ IoT Botnets to conduct DDoS attacks to target multiple devices simultaneously. According to Akhilesh et al., (2022), cyber intruders mainly target smart home appliances that have inadequate security measures to turn them into a Botnet to attack critical digital infrastructures.

Changing IoT devices into botnets to execute DDoS attacks involves several stages (Kumari & Jain, 2023), such as (1) identifying and obtaining access to IoT devices, (2) altering the device's code to accomplish malicious activities, and (3) directing the infected IoT devices and executing DDoS. Through this mechanism, gigantic false traffic can be created. For example, the Mirai Botnet instigated a massive DDOS attack in 2016, the largest one documented up to that point (Akhilesh et al., 2022). The study also elaborates that the DDoS attack exploited over 300,000 infected IoT devices, disrupting several significant websites and digital services, including GitHub, PayPal, Amazon, the BBC, PlayStation Network, and Spotify.

DDoS attacks leveraging IoT devices typically fall into volumetric attacks, protocol attacks, and application-layer attacks as tabulated in Table 1. Volumetric attacks involve overwhelming a network with excessive traffic, consuming all available bandwidth. Protocol attacks target weaknesses in network protocols, such as SYN floods and DNS amplification, to exhaust server resources. Application-layer attacks focus on disrupting services by sending malicious requests that overload servers, such as HTTP floods. Table 1 shows the details of types of DDoS attacks leveraging IoT devices.

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Table 1: Types Of DDoS Attacks Leveraging IoT Devices

| Type of Attack | Description | Impact |
|---|---|---|
| Volumetric Attacks | Floods a network with excessive traffic, consuming all available bandwidth and preventing legitimate communication. | Severely depletes network bandwidth, causing widespread service downtime. Affects IoT networks, cloud platforms, and enterprise infrastructure. |
| Protocol Attacks | Exploits vulnerabilities in network protocols, exhausting server resources by overwhelming connection requests and packet processing. | Disrupts network protocol operations, leading to server crashes and degraded network performance. Common in IoT botnet-based DDoS campaigns. |
| Application-Layer Attacks | Targets specific applications or services with malicious requests, overloading servers and disrupting normal operations. | Overloads web applications and online services, causing increased response times, degraded user experience, and financial losses due to downtime. |

Aside from that, hackers are creating more innovative and resilient ways that can hide their activity and spread rapidly. For example, hacker uses a node with a spoofed IP address to hide the real source of the attack (Sripriyanka & Mahendran, 2024). Another significant point to consider is that IoT devices become attractive to be targeted by various hacker levels, including non-technical users, due to several factors such as limited processing power, widespread deployment, and being placed at remote or unmonitored locations (W. Liu et al., 2024; Na et al., 2023; J. H. Park et al., 2024).

## 4.2 Blockchain Technology for IoT

Conventional security models rely on trusted third parties for authentication, access control, and data management, making IoT networks highly susceptible to single points of failure, unauthorized access, and large-scale cyberattacks (Wu et al., 2024). The need for a decentralized, tamper-proof, and trustless security framework has led to increased interest in blockchain technology as a potential solution (Lekssays et al., 2023; W. Liu et al., 2024).

Blockchain is a distributed ledger technology (DLT) that records transactions in a decentralized and immutable manner. It operates on the principles of transparency, cryptographic security, and consensus mechanisms, ensuring that data cannot be altered or deleted once recorded (Kadri et al., 2024). Unlike traditional client-server architectures, where a central authority controls network security, blockchain distributes control across multiple peer nodes, eliminating the risk of single-point failures and centralized breaches. This decentralized nature makes blockchain a promising candidate for enhancing security, data integrity, and access control in IoT networks (Alkhammash, 2024).

In addition to its tamper-resistant ledger, blockchain provides secure device authentication and end-to-end encryption for IoT communications. Each IoT device can be assigned a unique cryptographic identity, ensuring that only verified entities can participate in network transactions (Na et al., 2023). This is particularly crucial in IoT environments, where unauthorized devices or compromised nodes can be used to launch cyberattacks, exfiltrate sensitive data, or disrupt critical services. Furthermore, blockchain's ability to establish decentralized trust models eliminates the reliance on third-party authentication servers, reducing the risk of DDoS attacks that exploit centralized weaknesses (Wu et al., 2024).

One of the key advantages of blockchain in IoT security is its ability to automate security policies through smart contracts. Smart contracts are self-executing programs stored on the blockchain that enforce predefined rules, such as granting or revoking access, detecting anomalies, and mitigating threats in real-time (Patidar & Jain, 2024). For instance, an IoT

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network utilizing blockchain-based smart contracts can automatically blacklist malicious nodes or restrict traffic from suspicious IP addresses without human intervention, significantly enhancing network resilience against attacks (Y. Park & Kim, 2023).

Despite its potential, blockchain integration in IoT is not without challenges. Scalability remains a major concern, as most blockchain networks suffer from limited transaction throughput, high computational costs, and increased latency (Aguru & Erukala, 2024). Additionally, resource-constrained IoT devices may struggle with the energy-intensive requirements of traditional blockchain consensus mechanisms, such as Proof of Work (PoW). To address these limitations, researchers are exploring alternative consensus models, such as Proof of Stake (PoS), Byzantine Fault Tolerance (BFT), and lightweight cryptographic protocols to optimize blockchain's applicability for IoT (Wu et al., 2024).

Continued research into hybrid blockchain models, AI-driven threat intelligence, and energy-efficient cryptographic techniques is crucial for enhancing the reliability, efficiency, and scalability of blockchain-based IoT security frameworks. As blockchain technology matures, its integration with edge computing, federated learning, and adaptive consensus mechanisms will play a vital role in ensuring secure, autonomous, and resilient IoT networks capable of mitigating large-scale cyber threats, including DDoS attacks and unauthorized intrusions.

## 5. Blockchain-Based Defense Strategies Against DDoS in IoT

The security vulnerabilities in IoT networks make them prime targets for DDoS attacks, necessitating the adoption of blockchain-based defense mechanisms to enhance decentralized authentication and access control.

### 5.1 Decentralized Trust Models

Unlike traditional centralized authentication systems, blockchain leverages distributed trust models, smart contracts, and cryptographic methods to ensure tamper-resistant identity verification and secure access control. By decentralizing security operations, blockchain can mitigate single points of failure, preventing botnet formations and unauthorized intrusions (Arachchige, Branch, et al., 2024). However, despite its advantages, blockchain-based authentication still faces scalability and computational efficiency challenges, particularly for resource-constrained IoT devices.

One of the core principles of blockchain in IoT security is decentralized trust management, which eliminates the need for centralized certificate authorities (CAs). Traditional IoT authentication relies on third-party verification, which can be compromised or overloaded during DDoS attacks. Blockchain-based trust models enable peer-to-peer identity verification using consensus algorithms and cryptographic hashes, ensuring that only legitimate devices participate in the network (K. Xu et al., 2024).

Additionally, blockchain maintains an immutable record of authentication events, preventing attackers from tampering with access logs. Some recent studies have proposed multi-chain authentication mechanisms, where different blockchain networks collaborate to cross-verify IoT device identities, further enhancing resilience against identity spoofing attacks (X. Liu et al., 2024).

To address the limitations of single-layer authentication models, researchers have introduced multi-code trust frameworks, which integrate behavioral analysis, reputation scores, and anomaly detection mechanisms into blockchain-based access control. These frameworks use smart contracts to dynamically update trust scores based on device interactions, traffic behavior, and security policy compliance (Na et al., 2023).

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IoT devices with a history of suspicious activities, unusual data flows, or repeated authentication failures can be flagged and isolated before they compromise network security. Additionally, AI-powered reputation systems have been proposed to enhance trust computation, allowing adaptive security decisions based on real-time risk assessments (Saveetha et al., 2024).

### 5.2 Blockchain-Enabled Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS)

Federated Blockchain-Based Intrusion Detection Systems (FBIDS) combine federated learning (FL) and blockchain to create a distributed attack detection framework. Unlike traditional IDS models, which require centralized data storage, FBIDS enables collaborative AI model training across multiple IoT nodes while preserving privacy and security (Saveetha et al., 2024). By using blockchain as a secure ledger, federated learning models can share attack patterns across nodes without exposing sensitive raw data, ensuring tamper-proof collaboration (X. Liu et al., 2023). Moreover, blockchain-integrated FL has eliminated the risk of single-point failures, making it a scalable solution for decentralized anomaly detection (C. Xu et al., 2023).

A Blockchain-Enabled Intrusion Prevention System (IPS) enhances DDoS resilience by integrating decentralized traffic monitoring and anomaly detection mechanisms. Unlike conventional firewall-based IPS, which relies on rule-based filtering, blockchain-based IPS employs real-time blockchain consensus to detect and block malicious traffic (Khairuddin et al., 2024).

The system leverages distributed ledger analysis, where multiple nodes verify and validate incoming requests, making it difficult for attackers to manipulate or evade detection. Moreover, federated machine learning models have been integrated into blockchain-based IPS to improve attack detection rates, reducing false positives while enhancing real-time mitigation strategies (Okporokpo et al., 2025).

### 5.3 Anomaly Detection Analysis

AI-driven blockchain models significantly improve traffic anomaly detection by leveraging machine learning and deep learning techniques to identify suspicious patterns in real time. Neural network-based models trained on blockchain-stored network logs can continuously learn and adapt to evolving DDoS tactics (Hassan et al., 2024).

Recent studies have proposed federated gradient boosting techniques that integrate AI and blockchain consensus to filter malicious botnet traffic before it reaches its intended target (Na et al., 2023). AI-enhanced blockchain solutions offer adaptive, self-learning security models capable of responding to zero-day cyber threats in real-time.

The summary of blockchain-based defense strategies against DDoS in IoT is summarized in Table 2.

Table 2: Summary of Blockchain-Based Defense Strategies Against DDoS In IoT

| Defense Strategy | Description | Key Benefits | Key References |
|---|---|---|---|
| Decentralized Trust Models | Uses blockchain's distributed trust models, smart contracts, and cryptographic methods to ensure secure authentication and access control. Eliminates reliance on centralized | Prevents botnet formations, ensures tamper-resistant authentication, and enables | Arachchige et al. (2024), K. Xu et al. (2024), X. Liu et al. (2024) |

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| | | | |
|---|---|---|---|
| | certificate authorities, reducing single points of failure. | decentralized identity verification. | |
| Multi-Code Trust Frameworks | Enhances authentication with behavioral analysis, reputation scores, and anomaly detection mechanisms. Uses smart contracts to dynamically update trust scores based on device interactions and security compliance. | Flags and isolates suspicious IoT devices before they compromise network security, improves adaptive security decisions. | Na et al. (2023), Saveetha et al. (2024) |
| Blockchain-Enabled IDS and IPS | Integrates federated learning with blockchain to create a distributed IDS, enabling collaborative AI model training. Blockchain-enabled IPS utilizes real-time consensus mechanisms to detect and block malicious traffic. | Enhances attack detection while preserving privacy, eliminates risk of single-point failures, improves real-time mitigation. | Saveetha et al. (2024), X. Liu et al. (2023), C. Xu et al. (2023), Khairuddin et al. (2024), Okporokpo et al. (2025) |
| Traffic Pattern Analysis | Utilizes AI and deep learning techniques within blockchain to identify suspicious traffic patterns. Neural network-based models analyze blockchain-stored logs to improve real-time anomaly detection and threat mitigation. | Adapts to evolving DDoS tactics, enables proactive botnet traffic filtering, and enhances zero-day threat detection. | Hassan et al. (2024), Na et al. (2023) |

## 6. Challenges in Blockchain-based DDoS Detection in IoT

The increasing reliance on blockchain-based solutions for DDoS attack detection in IoT networks presents a promising yet challenging landscape. While blockchain offers decentralized security, immutable logs, and trustless collaboration, several key challenges delay its widespread adoption. Based on the thematic analysis of the provided research papers, we identify four primary challenges related to blockchain-based DDoS detection in IoT environments.

### 6.1 Scalability and Performance Bottlenecks

Scalability remains a significant barrier to the wide-scale adoption of blockchain in IoT due to its high transaction latency and resource-intensive operations. Blockchain systems inherently suffer from high computational overhead and latency issues, making them unsuitable for real-time DDoS detection in large-scale IoT environments.

The trust-based ensemble consensus incorporates PoW and PoS mechanisms to evaluate trust within the network, enhancing security by ensuring that only validated and authenticated nodes participate in transactions. This approach strengthens resilience against DDoS attacks by preventing malicious botnets from infiltrating the system and distributing network control across multiple nodes, reducing the risk of a single point of failure. However, a key drawback of this method is that as the number of nodes increases, network throughput gradually declines due to the computational overhead associated with PoW and PoS validation (Okporokpo et al., 2025). This limitation can become critical during large-scale DDoS attacks, where a surge in malicious traffic could slow down legitimate transactions, delaying attack detection and mitigation efforts.

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Blockchain's ability to detect and log attack traces in IoT networks is significantly impacted by on-chain storage limitations. As IoT devices generate massive amounts of data, storing all security-related events directly on the blockchain becomes inefficient and resource-intensive (W. Liu et al., 2024; Meng et al., 2023). Given the limited block sizes and transaction throughput of most blockchain architectures, the process of logging attack patterns, anomaly detections, and security alerts can lead to network congestion and delayed response times (Na et al., 2023; Yang et al., 2024).

This limitation reduces blockchain's efficiency in real-time DDoS detection and mitigation, as IoT networks struggle to store, verify, and process high volumes of security logs while maintaining decentralization and low latency (Saveetha et al., 2024). To address these challenges, hybrid storage models that integrate off-chain solutions such as cloud/fog/edge computing with blockchain architectures are being explored to enhance storage scalability and real-time attack detection (Katib & Ragab, 2023).

While blockchain is inherently decentralized, many existing distributed storage systems still rely on centralized gateways to efficiently process users' data storage and retrieval requests (Na et al., 2023). This centralization issue creates a single point of failure, making these gateways vulnerable to targeted DDoS attacks that can disrupt data availability and system reliability.

## 6.2 Resource Constraints in IoT Devices

The integration of blockchain technology into IoT networks has introduced scalability and efficiency challenges, particularly due to resource limitations in IoT devices. Many IoT nodes operate with limited processing power, memory, and energy resources, making it difficult to execute computationally intensive consensus mechanisms such as PoW and PoS (Arachchige, Branch, et al., 2024; Kumari & Jain, 2023).

Cryptographic operations such as hash functions, digital signatures, and smart contract execution impose heavy computational loads on IoT devices, further limiting their ability to participate in real-time blockchain transactions for DDoS detection (Kadri et al., 2024; Na et al., 2023). Additionally, Hyperledger Fabric, a widely used permissioned blockchain framework, faces storage and processing limitations, making it inefficient for lightweight IoT environments requiring real-time security monitoring (Patidar & Jain, 2024; K. Xu et al., 2024).

Moreover, blockchain's high communication overhead further intensifies resource constraints in IoT networks. Each IoT node participating in a blockchain network must validate, store, and broadcast transactions, leading to excessive bandwidth consumption and increased latency (Khairuddin et al., 2024; Saveetha et al., 2024). This issue is particularly problematic in low-power sensor networks, where constant synchronization with the blockchain ledger can drain over 60% of an IoT device's battery life, drastically reducing device operational longevity (Almadhor et al., 2024; Arachchige, Branch, et al., 2024).

Additionally, DDoS attacks targeting blockchain-integrated IoT systems can cause hardware failures and rapid energy depletion, further exacerbating existing computational constraints. During high-volume attack spikes, blockchain-integrated IoT nodes experience unusual CPU and memory consumption surges, leading to thermal stress, hardware degradation, and, in extreme cases, system shutdowns (Hassan et al., 2024; Kadri et al., 2024). Research has shown that DDoS-induced processing spikes in blockchain networks can cause IoT devices to overheat beyond 90°C, resulting in loss of security logs, increased attack vulnerability, and decreased overall system reliability (Kumari & Jain, 2023; K. Xu et al., 2024).

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To mitigate these resource constraints, researchers are exploring off-chain storage solutions, lightweight consensus mechanisms, and edge computing integration. By offloading transaction validation to edge nodes or cloud infrastructures, blockchain-enabled security systems can reduce the computational burden on IoT devices, allowing them to remain energy efficient while ensuring secure attack detection (Na et al., 2023; Patidar & Jain, 2024). Additionally, alternative consensus mechanisms such as Proof of Authority (PoA) and Proof of Reputation (PoR) are being explored to replace energy-intensive PoW, providing faster transaction validation with minimal computational costs (Almadhor et al., 2024; Saveetha et al., 2024).

### 6.3 Integration Challenges with Existing IoT Infrastructures

The integration of blockchain into existing IoT infrastructures for DDoS detection and mitigation presents several technical and operational challenges. Many IoT networks were originally designed with centralized architectures, relying on cloud-based security mechanisms, making it difficult to seamlessly incorporate decentralized blockchain-based security frameworks (Arachchige, Branch, et al., 2024; Kumari & Jain, 2023). Blockchain operates using distributed ledger technology (DLT), which requires all nodes to participate in consensus mechanisms and maintain copies of security logs, increasing computational and storage demands on IoT devices (Kadri et al., 2024; K. Xu et al., 2024).

Another key challenge is interoperability between blockchain frameworks and legacy IoT security models. Most IoT systems utilize lightweight security protocols that prioritize low processing power and energy efficiency, whereas blockchain consensus mechanisms such as PoW and PoS require significant computational and network resources (Khairuddin et al., 2024; Na et al., 2023). The lack of standardized blockchain-IoT communication interfaces complicates efforts to integrate existing IoT devices with blockchain-based security frameworks, leading to latency issues, inefficient DDoS detection, and increased overhead for data validation (Almadhor et al., 2024; Hassan et al., 2024).

Additionally, blockchain-enabled IoT security solutions require real-time attack detection and mitigation capabilities, yet many blockchain protocols have high transaction confirmation times. The use of public blockchains for logging and verifying DDoS attack signatures introduces scalability concerns, as IoT networks generate high-frequency security data that blockchains may struggle to process efficiently (Patidar & Jain, 2024; Saveetha et al., 2024). The challenge is further exacerbated in resource-constrained IoT environments, where real-time consensus processing may introduce security delays, reducing the effectiveness of blockchain-based anomaly detection models (Na et al., 2023; K. Xu et al., 2024).

To overcome these integration challenges, researchers are exploring cross-chain interoperability solutions, hybrid blockchain models, and AI-driven security frameworks. Cross-chain bridges and blockchain-agnostic APIs are being developed to enable communication between different blockchain protocols and IoT systems, facilitating real-time DDoS attack correlation across heterogeneous infrastructures (Patidar & Jain, 2024; K. Xu et al., 2024). Additionally, hybrid blockchain architectures that integrate permissioned and permissionless models provide customizable security configurations, ensuring that IoT environments can leverage blockchain's immutability without overwhelming device resources (Hassan et al., 2024; Na et al., 2023). Future research should focus on developing standardized security protocols and lightweight blockchain consensus mechanisms to facilitate blockchain adoption in large-scale IoT deployments.

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### 6.4 Standardization and Regulatory Considerations

The lack of standardization and regulatory frameworks presents a major challenge in the adoption of blockchain-based DDoS detection strategies in IoT networks. Unlike traditional cybersecurity solutions, blockchain-based security models lack universally accepted governance structures, leading to inconsistencies in implementation across different IoT ecosystems (Arachchige, Branch, et al., 2024; Kumari & Jain, 2023). The absence of global security protocols for blockchain in IoT complicates interoperability and compliance with industry standards, particularly in sectors that require strict security regulations, such as healthcare, finance, and critical infrastructure (Kadri et al., 2024; K. Xu et al., 2024).

One of the key regulatory challenges is the conflict between blockchain's immutability and existing data protection laws such as the General Data Protection Regulation (GDPR) and the California Consumer Privacy Act (CCPA). While blockchain's decentralized and immutable nature enhances DDoS resilience and attack traceability, it contradicts privacy requirements that mandate the ability to modify or delete personal data (Khairuddin et al., 2024; Na et al., 2023). Additionally, the legal status of blockchain-based evidence in cybercrime investigations remains unclear, as many jurisdictions have not yet established policies regarding the validity of blockchain-logged security events in forensic analysis (Almadhor et al., 2024; Hassan et al., 2024).

Another major concern is the lack of regulatory oversight and compliance frameworks for blockchain-powered security models in IoT networks. While traditional security solutions must adhere to compliance standards such as ISO/IEC 27001, NIST cybersecurity frameworks, and PCI-DSS, blockchain-based security lacks formal regulatory guidance, leading to uncertainty in enterprise adoption (Patidar & Jain, 2024; Saveetha et al., 2024). Without regulatory clarity, industries remain hesitant to integrate blockchain-based security models, fearing legal and operational risks associated with data integrity, liability, and jurisdictional disputes (Na et al., 2023; K. Xu et al., 2024).

Furthermore, cross-border legal discrepancies further hinder the scalability of blockchain-based IoT security solutions. Blockchain transactions occur globally, making it difficult to enforce unified cybersecurity policies across different regulatory jurisdictions. Some countries have adopted blockchain-friendly policies, while others restrict its use due to security and privacy concerns, making it challenging to establish cross-border DDoS defense networks (Kadri et al., 2024; Kumari & Jain, 2023). This regulatory fragmentation limits the ability of blockchain-based DDoS detection frameworks to function efficiently across international IoT deployments.

To address these standardization and regulatory concerns, policymakers and cybersecurity experts are advocating for blockchain governance models, privacy-preserving cryptographic techniques, and industry-wide security standards. The development of blockchain regulatory sandboxes, controlled testing environments that allow organizations to experiment with blockchain security solutions while ensuring compliance with evolving legal frameworks, has been proposed as a potential solution (Patidar & Jain, 2024; K. Xu et al., 2024). Additionally, integrating privacy-enhancing technologies such as zero-knowledge proofs (ZKPs), homomorphic encryption, and selective data anonymization into blockchain-based security frameworks could help align blockchain's security benefits with global data protection laws (Hassan et al., 2024; Na et al., 2023).

Table 3 provides a structured summary of these challenges, highlighting their impact, proposed solutions, and key references.

Table 3: Summary of Challenges in Blockchain-Based DDoS Detection in IoT

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| Challenge | Description | Proposed Solutions | Key References |
|---|---|---|---|
| Scalability and Performance Bottlenecks | Blockchain's high transaction latency and computational overhead hinder real-time DDoS detection in large-scale IoT environments. PoW and PoS mechanisms increase security but reduce network throughput as node count rises. On-chain storage limitations lead to congestion, reducing efficiency in detecting and mitigating DDoS attacks. | Hybrid storage models integrating off-chain solutions (cloud, fog, edge computing) to improve scalability. Alternative consensus mechanisms like PoA and PoR to reduce computational overhead. | Okporokpo et al. (2025), Liu et al. (2024), Meng et al. (2023), Na et al. (2023), Yang et al. (2024), Saveetha et al. (2024), Katib & Ragab (2023) |
| Resource Constraints in IoT Devices | IoT devices have limited processing power, memory, and energy, making it difficult to run resource-intensive blockchain consensus mechanisms like PoW and PoS. Cryptographic operations and high communication overhead drain battery life, reducing operational longevity. DDoS-induced processing spikes cause overheating and system failures. | Offloading transaction validation to edge/cloud infrastructures to reduce device resource consumption. Using lightweight consensus mechanisms and AI-based optimization to minimize energy usage while maintaining security. | Arachchige et al. (2024), Kumari & Jain (2023), Kadri et al. (2024), Na et al. (2023), Patidar & Jain (2024), Xu et al. (2024), Almadhor et al. (2024), Hassan et al. (2024) |
| Integration Challenges with Existing IoT Infrastructures | Most IoT infrastructures were designed for centralized security models, making blockchain integration challenging. Blockchain's distributed ledger and consensus mechanisms increase computational and storage demands on IoT devices. Lack of standard communication protocols results in latency issues and inefficiencies in DDoS detection. | Developing cross-chain interoperability solutions, hybrid blockchain models, and AI-driven security frameworks to facilitate blockchain-IoT integration. Implementing blockchain-agnostic APIs for better compatibility. | Arachchige et al. (2024), Kumari & Jain (2023), Kadri et al. (2024), Xu et al. (2024), Khairuddin et al. (2024), Na et al. (2023), Hassan et al. (2024) |
| Standardization and Regulatory Considerations | The absence of standardized security frameworks and regulatory oversight complicates blockchain adoption in IoT. Immutability conflicts with GDPR and CCPA data protection laws, making compliance difficult. Cross-border legal differences hinder large-scale deployments, and the lack of blockchain-specific compliance standards raises uncertainty in enterprise adoption. | Creating blockchain regulatory sandboxes for controlled testing and compliance verification. Adopting privacy-preserving cryptographic techniques like zero-knowledge proofs (ZKPs) to align blockchain with global data protection laws. | Arachchige et al. (2024), Kumari & Jain (2023), Kadri et al. (2024), Xu et al. (2024), Khairuddin et al. (2024), Na et al. (2023), Hassan et al. (2024) |

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## 7. Conclusion

This review successfully achieved its three primary objectives: (1) analyzing the impact of DDoS threats on IoT networks, (2) evaluating blockchain-based security frameworks for mitigating these attacks, and (3) identifying key challenges and future research directions. The study demonstrated that blockchain technology provides a decentralized, transparent, and tamper-resistant security solution that strengthens IoT network authentication, traffic monitoring, and anomaly detection. By leveraging distributed trust models, cryptographic authentication, and smart contracts, blockchain-based frameworks reduce reliance on centralized security models, mitigating single points of failure and improving attack response times.

However, the integration of blockchain in IoT security remains challenging due to high computational demands, scalability limitations, interoperability issues with legacy IoT infrastructures, and the lack of regulatory standardization. Addressing these challenges requires continued advancements in lightweight blockchain architectures, hybrid off-chain storage solutions, AI-enhanced security analytics, and standardized blockchain-IoT communication protocols. Future research should focus on developing more energy-efficient consensus mechanisms, integrating cross-chain interoperability frameworks, and refining blockchain governance models to facilitate widespread adoption in real-world IoT deployments.

By overcoming these challenges, blockchain-based security solutions have the potential to significantly enhance IoT resilience against sophisticated DDoS attacks, ensuring long-term network security, scalability, and adaptability in the evolving digital landscape.

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## CRediT authorship contribution statement

Fiza Abdul Rahim: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. Abdul Ghafar Jaafar: Writing – review & editing, Supervision, Validation, Methodology, Investigation. Noor Hafizah Hassan: Writing – review & editing, Supervision. Abdulrahman Aminu Ghali: Validation, Methodology, Investigation, Conceptualization. Khairol Shapawi Abdul Karim: Validation, Methodology, Investigation, Conceptualization.

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